

AUDITORY GROUPING DAILY

Tuesday, 07 July 2026 · 275 papers screened · curated

Today's batch is rich, with a throughline around prediction and sequence boundaries: a new MEG study dissociates fast predictive activity from slower representational reorganization in statistical learning (today's Paper of the Day), a somatosensory oddball study and a sharp critique from Nelken and Moshitch both bear on interpreting prediction-error signals in auditory/prefrontal recordings, and a birdsong study shows dopamine tagging a learned sequence's onset rather than its later portions. There's also a cluster on circuit implementation of latent states and context - orbitofrontal-striatal belief updating, motor-cortex 'contingency neurons', oxytocin-gated auditory-to-prefrontal connectivity - relevant to the mouse prelimbic/auditory-cortex recordings and the GLM's slow-drift module. Concrete tie for today: read the Nelken critique before claiming a distractor-chord response reflects genuine prediction rather than adaptation. Four of today's fourteen picks came from watch-author/citation follow-up rather than the deterministic feeds, and turned out to be the most methodologically load-bearing.

CHUNKING & SEQUENCE PROCESSING

Multiple event segmentation mechanisms in the human brain

RSS:ELIFE *eLife* — 2026-07-07

Corresponding: Jeffrey M. Zacks — Washington University in St. Louis — United States | [Open paper](#)

Using fMRI and two computational models - one generating event boundaries from prediction error, one from prediction uncertainty - the authors show human-identified event boundaries correspond to a specific temporal sequence of neural pattern changes (early shifts in anterior temporal cortex, then parietal cortex, then whole-brain pattern stabilization), with error-driven and uncertainty-driven boundaries leaving distinguishable neural signatures. This parallels the core theoretical distinction in the human chunk-in-stream task between bottom-up statistical-boundary detection and top-down template-matching detection of a memorized target sequence, and offers a concrete analysis template - fit an error-driven versus an uncertainty-driven boundary model, then look for dissociable pattern-shift signatures - that could be applied to high-gamma or single-unit data around Start-Target and End-Target boundaries in angular gyrus, supramarginal gyrus, or inferior/middle frontal gyrus.

Striatal Dopamine at Learned Sequence Boundaries Sustains Birdsong

BIORxIV *Xiao, L. et al. — bioRxiv — 2026-07-06*

Corresponding: Todd F. Roberts — University of Texas Southwestern Medical Center — United States | [Open paper](#)

In zebra finches, phasic dopamine transients in the song-related basal ganglia circuit shift from marking the later phases of a song sequence to marking sequence onset specifically as the song matures with practice, and optogenetically silencing dopamine right at sequence onset causes the learned song to gradually and severely deteriorate. This shows a sequence boundary - specifically the onset of a learned sequence - is itself a privileged, dopamine-tagged event during natural sequence learning, bearing directly on the mouse sequence-discrimination task's target-onset coding and on the GLM's trial-history module: sequence-initiation itself, not just outcome, may be a distinct dopaminergic signal worth looking for in prelimbic or auditory-cortex recordings around the first chord of a target trial.

AUDITORY & PREFRONTAL ELECTROPHYSIOLOGY

Oxytocin selectively biases sensory–prefrontal communication through network-level suppression and theta coupling

CROSSREF (MODEL-DISCOVERED) DaYoung Jung et al. — *Proceedings of the National Academy of Sciences* — 2026-06-09

Corresponding: Jee Hyun Choi — University of Science and Technology — South Korea | [Open paper](#)

Using multisite local field potential recordings across mouse auditory cortex and medial prefrontal cortex, the authors show oxytocin selectively boosts auditory-cortex and prefrontal responses to infant calls while suppressing baseline spontaneous activity (raising signal-to-noise), and specifically increases low-theta phase coupling and directional connectivity from auditory cortex to prefrontal cortex during rest. This is a directly relevant precedent for the prelimbic/auditory-cortex Neuropixels recordings in the mouse sequence task: it demonstrates using directional, Granger-causal connectivity measures between exactly these two regions to characterize how auditory information is gated into prefrontal cortex, and raises neuromodulatory state as a candidate source of session-to-session variability in auditory-to-prelimbic coupling.

BEHAVIORAL & COMPUTATIONAL MODELING

A Common Neural Signal of Evidence Accumulation for Perceptual and Mnemonic Decisions

BIORXIV Tsvinev, A. et al. — *bioRxiv* — 2026-07-06

Corresponding: Jason Samaha — University of California, Santa Cruz — United States | [Open paper](#)

The authors show the centro-parietal positivity, an EEG signature of evidence accumulation in perceptual decisions, also appears during memory-based decisions (semantic and episodic judgments) with a build-up rate that scales with evidence strength, response time, and confidence in both domains - arguing for a domain-general accumulate-to-bound signal rather than one specific to sensory evidence. This supports treating evidence accumulation in the leaky-accumulator drift-diffusion model family as a general computational motif rather than needing a separate account for each new evidence type (sensory chord identity versus sequence memory), and is a useful precedent for any future human EEG analogue of accumulation in the chunk-in-stream task's yes/no detection responses.

Motor implementation of context- and reward-based urgency regulation during action-oriented decision making.

PUBMED Thibault Fumery et al. — *Journal of Neuroscience* — 2026

Corresponding: Julie Duque — Université catholique de Louvain — Belgium | [Open paper](#)

Using single-pulse TMS to probe corticospinal excitability during a token-based decision task, the authors dissociate two sources of decision 'urgency' - task-context demands versus reward prospects - and show they are implemented through different motor-system mechanisms (broad leg-muscle excitability modulation versus hand-muscle surround inhibition), varying with individual impulsivity traits. This grounds the urgency/collapsing-bound component of the 24-model drift-diffusion family for the mouse sequence task: a context-driven and a reward-driven urgency signal may have genuinely distinct implementations rather than being a single unified urgency term, which could motivate splitting that toggle in a future model variant.

A corticostriatal circuit updates subjective beliefs about latent task states

CROSSREF (MODEL-DISCOVERED) Margaret L. DeMaegd et al. — 2026-03-14

Corresponding: Christine M. Constantinople — New York University — United States | [Open paper](#)

Recording from orbitofrontal-cortex neurons that project specifically to the caudate putamen in rats performing a hidden-reward-state task, the authors show this projection-specific population encodes categorical evidence for the current latent reward state, and perturbing it directly biases the rat's inferred belief about which state it's in. This is a strong precedent for the standing interest in continuous, non-stationary behavioral dynamics and specifically for the GLM's slow-drift module (satiety or time-on-task effects): a state-space-style latent belief has an identifiable, causally manipulable circuit locus, giving a concrete target class - projection-defined orbitofrontal-striatal populations - to consider beyond prelimbic cortex for where a mouse's slow engagement or strategy drift might be represented.

STATISTICAL LEARNING & ODDBALL

Learning regularities in noise engages both neural predictive activity and representational changes.

PUBMED Coumarane Tirou et al. — *Nature Communications* — 2026

Corresponding: Romain Quentin — Université Claude Bernard Lyon 1 — France | [Open paper](#)

Using MEG during a visuomotor statistical-learning task, the authors show that learning non-adjacent temporal dependencies unfolds in two dissociable stages: stimulus-specific predictive activity emerges rapidly and precedes any measurable behavioral improvement, followed only later by a slower increase in neural pattern similarity between the dependent, non-adjacent elements themselves. This maps onto the two components worth separating in the mouse sequence-discrimination task and its GLM strategy models: an early, fast anticipatory signal (the decision-window d' analysis) versus a slower representational reorganization of the sequence identity itself (the chunk-identity strategy module). It gives a concrete two-stage timeline to look for in prelimbic or auditory-cortex recordings as a target-sequence representation consolidates across training sessions, and reinforces that transitional-probability learning over non-adjacent elements has a genuine electrophysiological signature separable from simple behavioral change.

Does stimulus preceding negativity reflect predictions in a somatosensory roving paradigm?

PUBMED Gianluigi Giannini et al. — *Journal of Neuroscience* — 2026

Corresponding: Felix Blankenburg — Freie Universität Berlin — Germany | [Open paper](#)

In a somatosensory oddball ('roving') paradigm, the authors find a pre-stimulus anticipatory EEG signal (the stimulus-preceding negativity) that shrinks as a repeated stimulus train gets longer, tracking accumulated stimulus history before the next stimulus even arrives - distinct from the well-studied post-stimulus mismatch response. This is a clean precedent for treating anticipatory, pre-outcome activity as a marker of accumulated sequence history, which is exactly what the decision-window analysis in the mouse sequence-discrimination task is built to isolate (evidence accumulated before the last chord's onset, deliberately separated from reward-consumption licking). It's worth checking whether prelimbic or auditory-cortex recordings carry an analogous pre-chord anticipatory signal that scales with how many chords of the current run a mouse has already heard, independent of the post-chord evoked response.

How (not) to study deviance sensitivity and predictive coding in auditory cortex

CROSSREF (MODEL-DISCOVERED) Israel Nelken et al. — *Hearing Research* — 2026-07

Corresponding: Israel Nelken — Hebrew University of Jerusalem — Israel | [Open paper](#)

Nelken and Moshitch argue that the standard method for isolating genuine 'deviance sensitivity' or prediction-error signaling in auditory cortex - comparing a deviant tone's response in an oddball sequence to the same tone in a random or ordered control sequence - is not a reliable test: simple adaptation models (narrowly-tuned stimulus-specific adaptation, or network models with firing-rate adaptation and synaptic depression) can reproduce a 'deviant greater than control' pattern with no memory of currently valid regularities and no actual prediction-error computation. This is a direct methodological caution for framing chunk-boundary or novelty responses in the mouse sequence task as evidence of genuine prediction rather than stimulus-specific adaptation, and argues that any claim about auditory or prefrontal cortex 'detecting' a distractor or an out-of-position chord needs an explicit adaptation-model control, not just a deviant-versus-control contrast.

LANGUAGE, STROKE & BCI

Evidence from formal logical reasoning reveals that the language of thought is not natural language.

PUBMED Hope Kean et al. — *PNAS* — 2026

Corresponding: Evelina Fedorenko — Massachusetts Institute of Technology — United States | [Open paper](#)

Combining fMRI in healthy adults with behavioral testing of patients with severe aphasia from extensive damage to language brain areas, the authors find logical reasoning tasks do not engage the language network and that aphasic patients perform normally on logic tasks despite their linguistic impairment - evidence that inductive and deductive reasoning recruit the domain-general 'multiple demand' network rather than language machinery itself. This is a useful independent data point for framing the human chunk-in-stream participant's profile (severely impaired comprehension, largely spared repetition): it suggests general inductive/logical reasoning capacity may be dissociable from the comprehension deficit, relevant to deciding how much general-cognitive versus specifically linguistic capacity to assume is intact when designing statistical-learning versus template-matching conditions for this participant.

CIRCUITS, METHODS & POPULATION ANALYSIS

Anterior lateral motor cortex enables contextual decision-making via dynamic reconfiguration of local circuits

CROSSREF (MODEL-DISCOVERED) Jia Shen et al. — *Cell Reports* — 2026-06

Corresponding: Michael N. Shadlen — Columbia University — United States | [Open paper](#)

Recording with two-photon imaging in mouse anterior lateral motor cortex during a context-dependent odor-association task, the authors identify 'contingency neurons' with dual selectivity for context and choice, and show context information reconfigures local circuit routing from sensory input to motor output rather than being abstracted into a context-independent code. This is a useful precedent for the mouse sequence-discrimination task's distractor-context generalization problem - whether a licked response generalizes across different preceding-chord contexts - suggesting mixed context-by-choice selectivity in prelimbic cortex is worth looking for directly, rather than assuming a clean context-invariant sequence code, since that would predict different generalization failure patterns than a fully abstracted representation would.

Frontoparietal Hub Connectivity Integrates Information from Multiple Sources.

PUBMED Stephanie C Leach et al. — *Journal of Neuroscience* — 2026

Corresponding: Kai Hwang — University of Iowa — United States | [Open paper](#)

Using a model-based functional-connectivity approach in human fMRI, the authors show frontoparietal 'connector hub' regions modulate their coupling with sensory and motor regions differently depending on the specific computational variable at play - entropy (uncertainty before a choice) increases hub coupling with input/output regions, while the inferred task representation separately modulates hub-motor coupling. This gives a template for decomposing frontoparietal integration into distinct, computationally-defined coupling regimes rather than a single static property, relevant to the human chunk-in-stream work's inferior and middle frontal gyrus recordings if characterizing how top-down template information and bottom-up statistical evidence are routed differently through the same frontoparietal sites during detection.

An abstract relational map emerges in the human medial prefrontal cortex with consolidation.

PUBMED Alon B Baram et al. — *Current Biology* — 2026

Corresponding: Mona M. Garvert — Julius-Maximilians-Universität Würzburg — Germany | [Open paper](#)

Combining a graph-learning task with fMRI repetition suppression and representational similarity analysis, the authors find an abstract relational map of learned associations emerges in the medial prefrontal cortex only across two scanning sessions separated by several days, not within a single session - direct evidence that structural abstraction away from sensory detail is a slow, multi-day cortical process in humans, mirroring prior rodent reports. This bears on the timeline of the human chunk-in-stream participant's seven recording sessions: if template or chunk representations reorganize similarly across days, representational similarity analysis across sessions (rather than only within-session decoding) may be necessary to detect true structural learning in angular gyrus, supramarginal gyrus, or frontal recordings, especially on the top-down template-matching side of the task.

OTHER NOTABLE

Development of auditory and spontaneous movement responses to music over the first postnatal year.

PUBMED Trinh Nguyen et al. — *eLife* — 2026

Corresponding: Giacomo Novembre — Italian Institute of Technology — Italy | [Open paper](#)

Combining EEG and markerless pose-estimation kinematics in 79 infants aged 3, 6, and 12 months listening to music, the authors find enhanced auditory neural responses to structured versus shuffled music are present from 3 months, but complex, structured spontaneous movement to music only emerges by 12 months, with no age group showing tightly coordinated movement. This is an independent data point for the standing interest in treating spontaneous, continuous movement as real behavioral data rather than a nuisance to filter out (the same philosophy behind the GLM's slow-drift module), and the pose-estimation approach used here for quantifying unprompted movement is directly the kind of continuous-behavior-tracking method relevant to that standing interest, even outside the auditory-sequence-discrimination context specifically.

PAPER OF THE DAY

Learning regularities in noise engages both neural predictive activity and representational changes.

Coumarane Tirou et al. — Nature Communications — 2026 — new pick — abstract only

Corresponding: Romain Quentin — Université Claude Bernard Lyon 1 — France | [Open paper](#)

- Using MEG during a visuomotor task where participants implicitly learned non-adjacent temporal dependencies embedded in noise, the authors dissociate two temporally distinct learning mechanisms rather than treating 'statistical learning' as a single process.
- A fast, stimulus-specific predictive signal appears before stimulus onset and precedes any measurable behavioral improvement, showing the brain can encode an upcoming regularity before that knowledge shows up in behavior.
- Only afterward does a slower process build up: increased neural pattern similarity specifically between the non-adjacent elements that are statistically linked - a genuine representational reorganization of the sequence structure itself, not just a strengthening of the early predictive signal.
- Both processes are carried by a distributed set of networks including sensorimotor and dorsal attentional regions rather than one dedicated 'statistical learning area', arguing against a single, localized neural signature of sequence learning - worth keeping in mind before searching for a single region that 'does' sequence learning in prelimbic or auditory cortex.

HOW TO READ IT (~20 min)

Full text wasn't retrievable, so budget the 20 minutes around the abstract plus whatever figures the published PDF makes available. Spend the first ~8 minutes on the Results, specifically the two analyses that produce the paper's central claim: the pre-stimulus decoding analysis that shows predictive activity emerging before behavioral improvement, and the representational-similarity analysis that shows increased pattern similarity between non-adjacent, statistically-linked elements later in learning - these are the two concrete methods worth understanding well enough to consider replicating in spike-sorted data. Spend ~5 minutes on Methods, focusing on exactly how 'non-adjacent temporal dependencies embedded in noise' were constructed in the visuomotor task, since that is the closest published methodological parallel to the mouse task's own non-adjacent generalization problem (BC = XBC = XXBC). Skim the Introduction and Discussion (~7 minutes combined), looking specifically for any claim about why the predictive signal precedes the representational one mechanistically (the abstract only asserts the ordering, not a mechanism) - if the Discussion speculates on this, that speculation is the part most worth stress-testing against prelimbic/auditory-cortex data.

YOUR QUESTION TO ANSWER

The BC-versus-CB task can only really move forward once it's known whether a mouse's licked response is riding on an early, fast anticipatory signal (paralleling this paper's stimulus-specific predictive activity) or on a slower, consolidated sequence-identity representation (paralleling their non-adjacent representational-similarity buildup) - if forced to bet which one the GLM's chunk-identity strategy weight is actually capturing at the point a mouse first reaches criterion on a new distractor stage, which would it be, and does the existing decision-window versus response-window split already allow testing this, or would a new within-session early-versus-late analysis be needed to separate them?